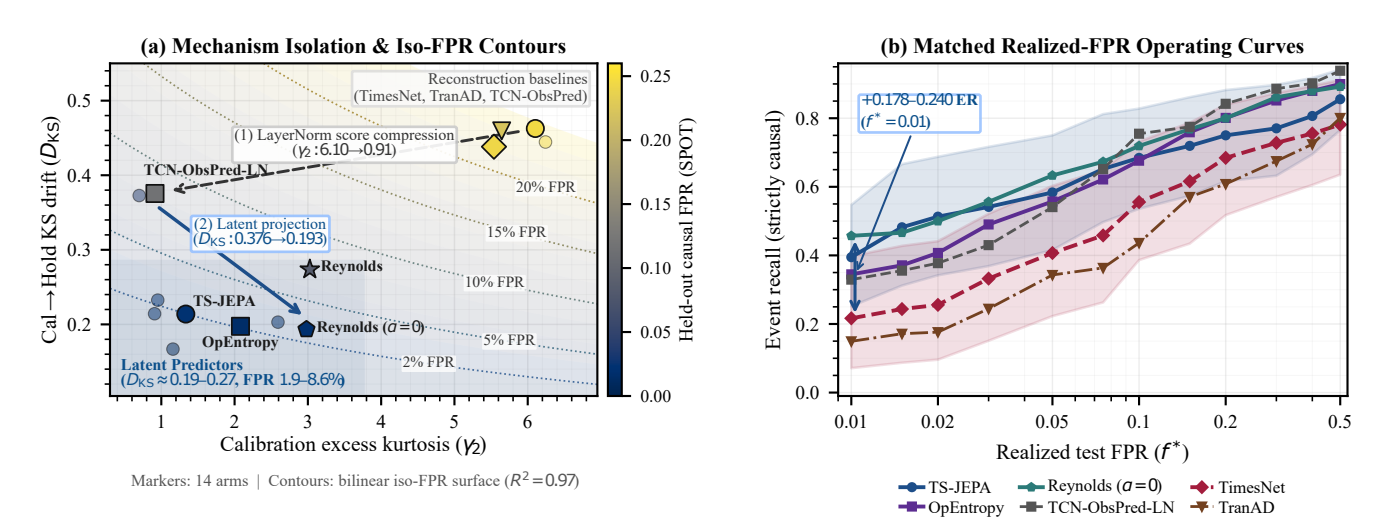

(a) Mechanism Isolation & Iso-FPR Contours
Reconstruction baselines
(TimesNet, TranAD, TCN-ObsPred)
(1) LayerNorm score compression
(γ_2 : 6.10→0.91)
TCN-ObsPred-LN
(2) Latent projection
(D_{KS} : 0.376→0.193)
Reynolds
TS-JEPA
Reynolds (σ=0)
OpEntropy
Latent Predictors
($D_{KS} \approx 0.19$–0.27, FPR 1.9–8.6%)
20% FPR
15% FPR
10% FPR
5% FPR
2% FPR
Cal→Hold KS drift (D_{KS})
Calibration excess kurtosis (γ_2)
Held-out causal FPR (SPOT)
Markers: 14 arms | Contours: bilinear iso-FPR surface (R^2=0.97)
(b) Matched Realized-FPR Operating Curves
+0.178–0.240 ER
(f^* = 0.01)
Event recall (strictly causal)
Realized test FPR (f^*)
TS-JEPA
Reynolds (σ=0)
TimesNet
OpEntropy
TCN-ObsPred-LN
TranAD